AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT131
Discrete Mathematics

Final Examination
Spring Semester 2025/2026

Date: 15 /June/2026

Number of Exam Pages: 4 pages (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **50**
- This exam consists of **5** questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×2 marks] Choose the correct answer:

1.1: The proposition $(P \longrightarrow Q) \wedge (Q \longrightarrow P)$ is

a) Tautology
b) Contradiction
c) Contingency
d) All of the above
e) None of the above

1.2: If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then

a) $A = B$
b) $A = C$
c) $B = C$
d) $AB = \emptyset$
e) None of the above

1.3: The decimal representation of $(101011)_2$ is

a) 43
b) 16
c) 32
d) 64
e) None of the above

1.4: Assume that R is a relation on $\{1, 2, 3, 4\}$ defined as follows: aRb if and only if $a + b$ is an odd number, then $R =$

a) $\{(1,2), (1,4), (2,3), (3,4)\}$
b) $\{(2,1), (4,1), (3,2), (4,3)\}$
c) $\{(1,2), (3,4)\}$
d) $\{(1,2), (2,1), (1,4), (4,1), (2,3), (3,2), (3,4), (4,3)\}$
e) None of the above

1.5: The number of elements in the adjacency matrix of a graph having 7 vertices is

a) 7
b) 14
c) 36
d) 49
e) None of the above

Each question is worth 10 marks. Answer the following questions:

Q–2:

a) [5 marks] Using truth tables show that $(p \longrightarrow q) \wedge (q \longrightarrow r) \longrightarrow (p \longrightarrow r)$ is a tautology.

b) [2 marks] Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z where $x = 0,1, or\ 2, y = 0\ or\ 1$, and $z = 0\ or\ 1$.Write out this propositions $\exists x Q(x, 1,1)$ using disjunctions and conjunctions.

c) [1 mark each] Find each of the following:

i. 5 mod 5.
ii. -75 div 13.
iii. $\gcd(64,512)$.

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Q–3:

a) **[2.5+2.5 marks]**
   i. The manager of a department store chain wishes to make a four-digit identification cards for the employees. How many different cards can be made if the digits 1, 2, 3, 4, 5, and 6 are used and repetitions are not allowed?
   ii. A newspaper editor has received 8 books to review. He decides that he can use 3 reviews in this newspaper. How many different ways can these reviews be selected?

b) **[2.5+2.5 marks]** Suppose that two integers from the set of 8 integers $\{1, 2, \ldots, 8\}$ are choosen at random. Find the probability that
   i. Both numbers match.
   ii. Sum of the two numbers picked is less than 4.

Q–4: Let $A = \{a, b, c, d\}$ and $R = \{(a, a), (a, b), (b, a), (c, c), (c, d), (d, c)\}$ be a relation on A.

a) **[2 marks]** Represent the given relation as a digraph.
b) **[2 marks]** Find the matrix representation of R.
c) **[2 marks]** Is R antisymmetric, transitive? Explain.
d) **[2 marks]** Calculate $R \circ R$.
e) **[2 marks]** Let R be the partial order relation defined on $A = \{2, 3, 4, 5, 6, 8, 10, 40\}$, where xRy means $x \mid y$. Draw the Hasse diagram for R.

Q–5:

a) **[3 +3 marks]** Given the adjacency matrix of a graph $G = (V, E)$ with vertices a, b, c, d:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

where alphabetical order is used to determine the rows and columns of the adjacency matrix.
   i. Draw the graph G.
   ii. Find the incident matrix of G.

b) **[4 marks]** Set up a binary search tree to sort the following numbers:
22, 8, 14, 17,3,9,27,11

End of questions

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